

The empirical recurrence for OEIS A281834, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A281834 records “Empirical recurrence of order 97” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 977$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 100$. Its last coefficient is $c_{97} = 13162496 \neq 0$, so no further tail satisfies anything shorter; any order-97 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A281834 is “Number of $n \times 5$ 0..1 arrays with no element equal to more than four of its king-move neighbors and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

16, 472, 5315, 74240, 1053407, 14783996, 208837689, 2957076233, ...

The entry states:

Empirical recurrence of order 97 (see link above).

As of the “Last modified” line on the live entry (Apr 26 2021 19:49:26, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 977$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 977$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 97: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 97.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1954$ exact terms from $n = 3$ on, it returns order 97 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{97} c_i a(n-i),$$

c_1	14	c_{26}	−983887746490	c_{51}	−496613258964538	c_{76}	7386496892286
c_2	5	c_{27}	301956336127	c_{52}	−729093781193888	c_{77}	9959180742
c_3	173	c_{28}	5810482204747	c_{53}	86956826666495	c_{78}	−2740040353351
c_4	−2492	c_{29}	8726385657039	c_{54}	−137019899327920	c_{79}	2541839020501
c_5	−7576	c_{30}	8961713860846	c_{55}	588519251867488	c_{80}	−3078358103449
c_6	−11212	c_{31}	−7277119332509	c_{56}	91553798284039	c_{81}	1367602687783
c_7	202524	c_{32}	−23267755976958	c_{57}	257335275984176	c_{82}	−871684915550
c_8	607052	c_{33}	−31837921688232	c_{58}	−397428278578227	c_{83}	576787946341
c_9	322699	c_{34}	−11556674986797	c_{59}	−183726651803170	c_{84}	108741574422
c_{10}	−7821984	c_{35}	44775396696817	c_{60}	90165336468062	c_{85}	−1884352646
c_{11}	−22462308	c_{36}	29099846110647	c_{61}	296652827534244	c_{86}	37770684815
c_{12}	−4665752	c_{37}	45232907064828	c_{62}	75939067246716	c_{87}	−75805444044
c_{13}	168332725	c_{38}	−95952020628954	c_{63}	−41940449129144	c_{88}	34535797790
c_{14}	535627235	c_{39}	−6308287969353	c_{64}	−254708143790806	c_{89}	−618291946
c_{15}	229459241	c_{40}	−34871643539358	c_{65}	21630407487445	c_{90}	13043878316
c_{16}	−1911943314	c_{41}	248781128003293	c_{66}	−69274394348601	c_{91}	−1731219148
c_{17}	−7397482253	c_{42}	49614064053574	c_{67}	85651993008709	c_{92}	−1094692368
c_{18}	−7160473466	c_{43}	−33607524518189	c_{68}	75043601049146	c_{93}	−810000064
c_{19}	6794031172	c_{44}	−424915382065765	c_{69}	−28951872251086	c_{94}	−155476224
c_{20}	54433254395	c_{45}	−323556594102862	c_{70}	54142131571786	c_{95}	111081408
c_{21}	95422165495	c_{46}	−105075987577053	c_{71}	−65764670889327	c_{96}	22648704
c_{22}	109509197643	c_{47}	472186756411509	c_{72}	17444269744721	c_{97}	13162496
c_{23}	−155369556225	c_{48}	659977469956569	c_{73}	−34991094516934		
c_{24}	−525566010337	c_{49}	566417746493016	c_{74}	5911015578997		
c_{25}	−1470926448403	c_{50}	−341465316072124	c_{75}	1978079161905		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 100$, and it is the recurrence OEIS A281834 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{97} - c_1x^{97-1} - \dots - c_{97}$. Its constant term is $-c_{97} = -13162496$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 97 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 97, so $\deg q = 97$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 97, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 97, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 13162496.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-97 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A281834.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.